

Unit II

Number Theory

classmate

Date _____

Page _____

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I * Divisibility:

* Division:

if a & b are integers with $a \neq 0$
then

a divides b

if there exists an integer c

such that $b = ac \Rightarrow c = b/a$

$\rightarrow$ The notation $\frac{a}{b}$ denotes that a divides b .

$\rightarrow$ if $\frac{a}{b}$ then $\frac{b}{a}$ is an integer

$\rightarrow$ if a does not divide b , we write $a \nmid b$

* Properties of Divisibility:

Theorem 1: let a, b and c are integers, where $a \neq 0$

1. if a/b and a/c then $a/(b+c)$

2. if a/b , then $\frac{a}{bc}$ for all integers c

3. If a/b and b/c then a/c

Example : 12 is divisible by 3

$$12 = 3 \times 4$$

2. Modulo Operator

The modulo operator finds the remainder when one number is divided by another

Notation $a \text{ Mod } n$

$$(a+b) \text{ mod } n = [(a \text{ mod } n) + (b \text{ mod } n)] \text{ mod } n$$

$$(a-b) \text{ mod } n = [(a \text{ mod } n) - (b \text{ mod } n)] \text{ mod } n$$

$$(a*b) \text{ mod } n = [(a \text{ mod } n) * (b \text{ mod } n)] \text{ mod } n$$

Example $7 \text{ mod } 3$ equals 1 (remainder 1)

$$= 7 \% 3 = 1$$

$$=$$

$$\begin{array}{r} 2 \\ 3 \overline{) 7} \\ \underline{6} \end{array}$$

$$\text{Example 2} = 10 \text{ mod } 3 = 1$$

$$= 1$$

$$\begin{array}{r} 3 \\ 3 \overline{) 10} \\ \underline{9} \end{array}$$

1 remainder.

$$= 15 \text{ mod } 4$$

$$\begin{array}{r} 3 \\ 4 \overline{) 15} \\ \underline{12} \end{array}$$

3 remainder

3. Set of residues $\mathbb{Z}_n$

→ The set of residues modulo n is

$$\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}.$$

→ Additive inverse of a is b such that
 $a + b = 0 \pmod{n}$

→ Multiplicative inverse of a is b such that
 $a \cdot b = 1 \pmod{n}$

→ When we are working in modular arithmetic we often ~~need~~ ^{need} to find inverse of a number relative to a operation

① Additive inverse

② Multiplicative "

$$(1) \quad a + b \equiv 0 \pmod{n} \Rightarrow (a + b) \pmod{n} = 0$$

Example: If $\mathbb{Z}_n$ additive inverse of a can be calculated as $\boxed{b = \mathbb{Z}_n - a}$

$$\text{e.g. } \mathbb{Z}_n = 10, a = 4$$

$$b = 10 - 4 = 6$$

$$\text{check} = (4 + 6) \pmod{10} =$$

$$= 10 \pmod{10}$$

$$= 0$$

Note = Always additive inverse exists on a set and inverse is unique

- each number has only one addl. inverse
- The inverse of the number may be the number itself.

Multiplicative Inverse

Two numbers a & b are the multiplicative inverse of each other if

$$(a \times b) \equiv 1 \pmod{n}$$

Example if $n=10$ & $a=3$

then $b=7$

$$(b = zn - a)$$

$$b = 10 - 3 \\ = 7$$

$$(a \times b) \equiv 1 \pmod{n}$$

$$= (3 \times 7) \pmod{10}$$

$$= 21 \pmod{10}$$

$$= 1 \quad (\text{remainder})$$

$$\begin{array}{r} 2 \\ 10 \overline{) 20} \\ \underline{20} \\ 0 \end{array}$$

1

* Calculation of GCD & Multiplicative Inverse

GCD = Greatest Common Divisor

→ The multiplicative inverse of a number a modulo m is another x such that $(a \times x) \% m = 1$

→ The multiplicative inverse exists only when a and m are co-prime (i.e. their GCD is 1)

Example: Find GCD of 12 & 18

$$12 = 6 \times 2 + 0$$

$$18 = 12 \times 1 + 6$$

$$\text{GCD}(12, 18) = 6$$

$$12 = 2 \times 2 \times 3$$

$$18 = 2 \times 3 \times 3$$

$$\text{Common Factors} = 2 \times 3 = 6$$

Q = Find GCD (36, 60)

$$36 = 2 \times 2 \times 3 \times 3$$

$$60 = 2 \times 2 \times 3 \times 5$$

GCD = Multiplication of common factors

$$= 2 \times 2 \times 3$$

$$= 12$$

$$\text{GCD}(36, 60) = 12$$

Euclidean Algorithm

→ The Euclidean algo based on the idea that $\text{gcd}(a, b)$ is equal to $\text{gcd}(a, c)$ when $a > b$ and c is the remainder when a is divided by b .

Example: Find $\text{gcd}(91, 287)$

$$287 = 91 \times 3 + 14$$

divide 287 by 91

$$91 = 14 \times 6 + 7$$

||

91 by 14

$$14 = 7 \times 2 + 0$$

||

14 by 7

$$\text{gcd}(287, 91) = \text{gcd}(91, 14) = \text{gcd}(14, 7) = 7$$

* Applications

- Modular arithmetic & inverses are central in RSA
- Diffie-Hellman Key exchange
- Digital signatures.